



DAT610: Ethics & Privacy

Overview

For this assignment, you will be the data scientist tasked with analysing a synthetic dataset to validate it before it can be approved for use in a fraud-detection model. Throughout, you will be working with two sets of data: a real dataset and a synthetic dataset that was created to complement the real dataset. Your job is to assess, within the context of the five-level validation framework iterated in class, whether the synthetic data set is safe and/or statistically valid for use, and then provide the supporting evidence that will support your recommendation.

Do you approve this synthetic data for release into production?

Tasks

You should come up with a dataset for fraud detection (> 1500 records) and generate synthetic data with a Conditional Tabular GAN (CTGAN) deep learning model. Thereafter, you validate the CTGAN dataset using the metrics listed on the slide.

Submission Report

Part 1: In your report (minimum of two pages), you should explain your workflow, the results you got with your dataset, with respect to every validation metric introduced in class. Then discuss your findings.

Part 2: A compressed folder of your Python script(s) with proper comments explaining the code and demarcating aspects of the exercise.

Note: There should be two sets of datasets: a 'real' dataset and a synthetic dataset for this exercise.